

Experiment 6

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Course Code: MLA0305

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Slot: B

Title: Monte Carlo Prediction and Control for Reinforcement Learning

Software Required:

- Python
- Gymnasium

AIM

To implement Monte Carlo prediction and control methods for Reinforcement Learning.

THEORY

Monte Carlo methods estimate state or action values using the actual returns obtained from completed episodes. They learn directly from experience without requiring a model of the environment.

PROCEDURE

1. Open the Python environment.
2. Import Python libraries and Gymnasium.
3. Create the required environment.
4. Initialize the policy and value estimates.
5. Start an episode from an initial state.
6. Generate actions according to the current policy.
7. Record the states, actions and rewards.
8. Calculate the return after completing the episode.
9. Update the value estimates and improve the policy.
10. Repeat the process for several episodes and observe the learning performance.

CODE:

```
import numpy as np
import matplotlib.pyplot as plt

# Sample episode rewards
episodes = [
    [1, 2, 3],
    [2, 1, 4],
    [3, 2, 5],
    [4, 3, 6],
    [5, 4, 7]
]

gamma = 0.9
returns = []

# Monte Carlo Return Calculation
for episode in episodes:
    G = 0
    for reward in reversed(episode):
        G = reward + gamma * G
    returns.append(G)

# Estimated State Value
state_value = np.mean(returns)

print("Monte Carlo Returns")
for i, r in enumerate(returns):
    print(f"Episode {i+1}: {r:.2f}")

print("\nEstimated State Value:", round(state_value, 2))

# Comparison Graph
plt.figure(figsize=(7,5))
plt.plot(range(1, len(returns)+1), returns, marker='o', label='Episode Return')
plt.axhline(state_value, linestyle='--', label='Average Return')

plt.title("Monte Carlo Prediction")
plt.xlabel("Episodes")
plt.ylabel("Return")
plt.legend()
```

```
plt.grid(True)  
plt.show()
```

OUTPUT:

Monte Carlo Returns

Episode 1: 5.23

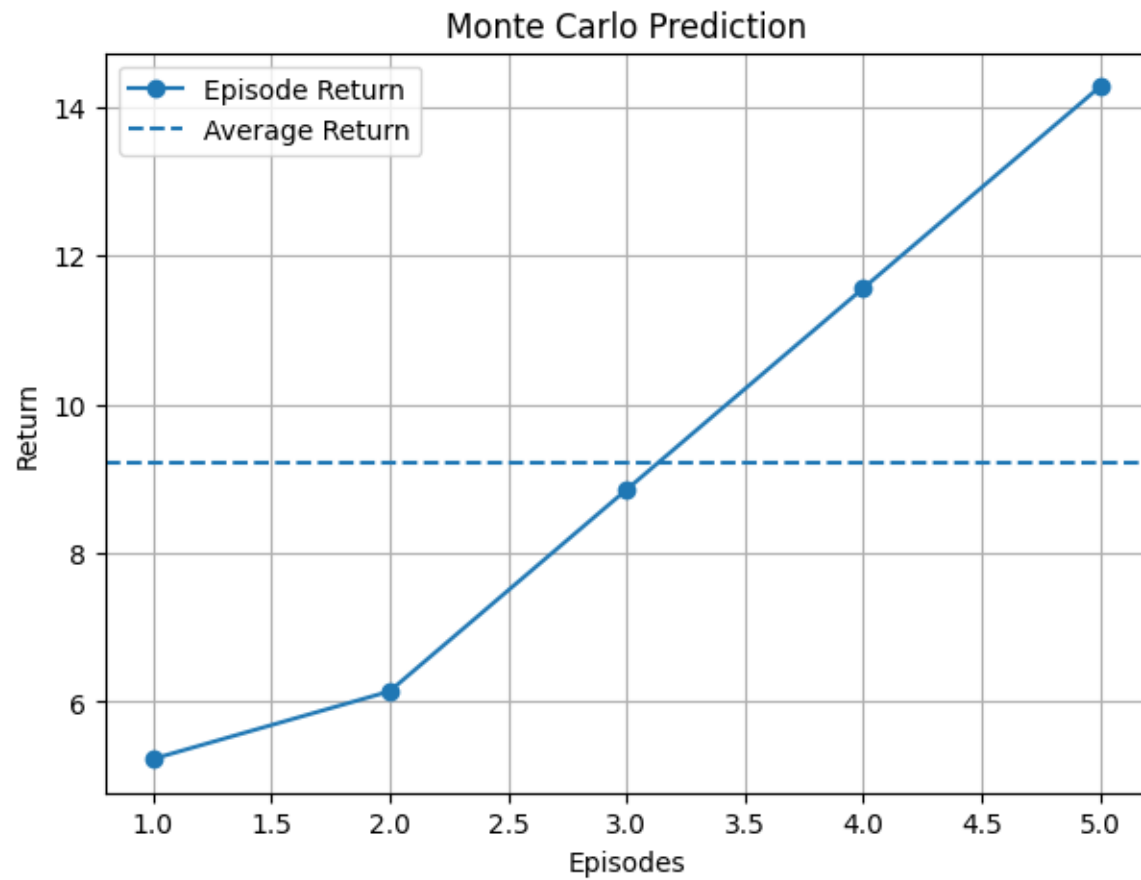
Episode 2: 6.14

Episode 3: 8.85

Episode 4: 11.56

Episode 5: 14.27

Estimated State Value: 9.21



RESULT

Monte Carlo prediction and control were successfully implemented for Reinforcement Learning.